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Do non-farmers pay more for land than farmers?

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Abstract

The increase in farmland prices in many parts of the world over the past decade has sparked discussions about whether non-farmers pay higher prices for farmland. This study uses a causal machine learning approach with a rich data set of land transactions in Germany to quantify the potential price premium paid by non-farmers. By applying the causal forest method, we uncover the heterogeneity of price premiums and reveal moderating effects of covariates. We find that the average positive price premium by non-farmers decreases with parcel size and distance to a highway exit.

Keywords: causal machine learning, causal forest, heterogeneous treatment effect, farmland, non-agricultural investors

JEL classification: C14; C55; Q15

1. Introduction

The pronounced price surge in agricultural land markets in many parts of the world during the last decade has triggered extensive empirical work that aims at identifying causes for the increase in land prices. There is a consensus that there is not a single factor, but instead there are multiple factors that have fueled this development, such as low cost of capital (Plogmann *et al*., 2020), increased land demand for energy production (Myrna, Odening and Ritter, 2019) and fierce competition among farmers for the fixed production factor 'land' (Cotteleer, Gardebroek and Luijt, 2008). Yet, one variable stands at the epicentre of the political discourse—the increased intrusion of non-farmers in the wake of the 2007 financial crisis (Byerlee *et al*., 2010; van der Ploeg, Franco and Borras, 2015). It is hypothesised that depressed returns on financial

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assets diverted financial investors' liquidity towards agricultural land markets, precipitating a price surge and effectively pricing farmers out of the market. Regrettably, empirical evidence either validating or refuting this supposition is scant.

This literature gap can be explained by two reasons. First, agricultural land prices are usually influenced by many determinants, which are typically summarised in rather complex hedonic price models. Second, the group of non-farmers is heterogeneous regarding legal form, size, motivations and region of provenance. Unfortunately, transaction data that include information about price determinants as well as buyer characteristics which would enable a comprehensive microstructural analysis of non-farmers' price impacts are rare. Nevertheless, some attempts have been made to assess the role of non-agricultural investors in land price formation. Hüttel, Wildermann and Croonenbroeck (2016) and Croonenbroeck, Odening and Hüttel (2020) find only minor price markups of foreign (non-German) investors in land privatisation auctions. Vyn and Shang (2021) identify higher prices for non-farmers in Canada, but only in near-urban areas. While Witzke *et al*. (2020) find no systematic differences between non-agricultural and agricultural investors by analysing German land auctions, Meißner and Mußhoff (2022) find lower sales prices for non-farmers and a considerable share of transactions with non-farmers as buyers. Another study that comes closest to this approach is by Curtiss *et al*. (2021) and uses transaction data from the Czech Republic to conduct a microstructural analysis that examines systematic differences in willingness to pay (WTP) for agricultural land between farmers and non-farmers, their heterogeneity and changes over time. It shows that while non-farmers initially pay higher prices than farmers, sales prices converge on average, with the speed of convergence influenced by factors such as non-farmer activity, overall land demand, seller structure and farm structure.

Against this backdrop, this article investigates the following research questions. First, we aim to ascertain whether there are systematic and potentially heterogenous differences in the WTP for agricultural land between farmers and non-farmers. We consider the purchase of a land plot by a non-farmer as a treatment and determine the average treatment effect (ATE) on land prices for a large set of land transactions in Lower Saxony (Germany). Second, we explore the possibility that these differences are subject to moderation by certain plot characteristics, namely parcel size, distance to the nearest highway exit and soil quality. Finally, acknowledging the multifaceted nature of land transactions, we investigate not-yet-identified factors that may have a moderating effect on these systematic differences.

Our study aims to address these gaps in the literature, contribute significantly to the existing body of knowledge and present novel methodological approaches regarding the examination of agricultural land markets. First, we take a pioneering step by applying the causal forest approach to agricultural land markets, utilising observed data. This machine learning method provides a more nuanced understanding of land price determination, allowing for the identification of heterogeneous treatment effects not typically discerned using

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traditional methods. Second, we further examine the ATE by decomposing it and assessing heterogeneity and interactions between the treatment effect and sales characteristics, thereby providing a more comprehensive understanding of price formation in land markets. This approach supplements the existing literature that typically centres around average effects. Lastly, our research provides an adaptable methodological framework for crafting more individualised and context-specific policies by demonstrating treatment effect heterogeneity. By revealing differential effects based on various plot characteristics, our results support the design of policies that consider this heterogeneity.

The structure of this article is as follows. Section 2 discusses the background and conceptual framework of WTP for agricultural land, examining differences between farmers and non-farmers. Section 3 outlines the empirical framework and approach to estimating heterogeneous treatment effects using causal forests. Information on the study region and data is provided in Section 4. Section 5 discusses the identifying assumptions of the model. Section 6 presents and discusses the results of the empirical application, and the final section concludes.

2. Background and conceptual framework

An assessment of the impact of the buyer type on price formation in farmland markets should account for peculiarities of these markets. Farmland is extremely heterogeneous in its attributes; it is immobile and non-duplicable. As a result, local farmland markets are typically thin with only a few potential buyers and sellers. In view of these peculiarities, Balmann *et al.* (2021) characterise farmland markets as search and bargaining markets opposed to homogeneous goods markets. In these markets, prices are the outcome of a negotiation process between a seller and buyer or of a bidding procedure in an auction-like format. In either case, the resulting transaction price reflects the WTP of the successful buyer, which must meet the willingness to accept (WTA) or the reservation price of the seller (Figure 1). Unlike in fully competitive homogeneous goods markets, WTP and WTA do not necessarily match, but rents can arise which are shared according to the relative bargaining power of the seller and buyer (Harding, Knight and Sirmans, 2003; Cotteleer, Gardebroek and Luijt, 2008; Curtiss *et al.*, 2021). The buyer's WTP, in turn, can be understood as the present value of expected future returns of land utilisation, i.e.

$$E\left(\sum_{t=1}^{t^*} R_t^A \cdot q^{-t} + \sum_{t=t^*+1}^{\infty} R_t^{NA} \cdot q^{-t}\right), \tag{1}$$

where R^A and R^{NA} denote returns from agricultural and non-agricultural land use, respectively, t^* is the switching time, q^{-t} is a discount factor, and E denotes an expectation operator. The notion of an expected present value is very general as it captures all kinds of land use and future development options of farmland. It also accounts for differences in cost structures among potential

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buyers, economies of size, managerial capabilities and inefficiencies. Moreover, it reflects the expectations about future returns, which may differ among buyers even for otherwise equal farming practices, i.e. a high price for a land plot may be the result of a buyer's overly optimistic beliefs.

The research focus of this paper boils down to the question as to whether systematic differences exist between farmers and non-farmers regarding their WTP for farmland. Land amenities and value components are most likely either appraised in a similar way by all buyers or are not related to the buyer type. The non-dispersion of land plots or eligibility for subsidies are examples of this type of land price determinants. In what follows, we discuss factors that may cause a gap in the WTP of farmers and non-farmers, as well as examine land traits for which the valuation may differ between buyer groups as portrayed in Figure 1. One has to acknowledge that the group of non-farmers itself is heterogeneous. Hüttel *et al.* (2013) differentiate non-agricultural investors according to their general economic background, legal form and remoteness from the land plot of interest. The motivation, intended land use, and thus WTP will differ within subgroups, so not all arguments provided below will apply to the same extent to all subgroups. Figure 1 also illustrates that it is impossible to directly infer from observed land prices the latent WTP and WTA. Observed land prices are not uniquely determined by WTP and WTA. Instead, they are the outcome of a negation or bidding process, which is also influenced by the relative bargaining power of market participants.

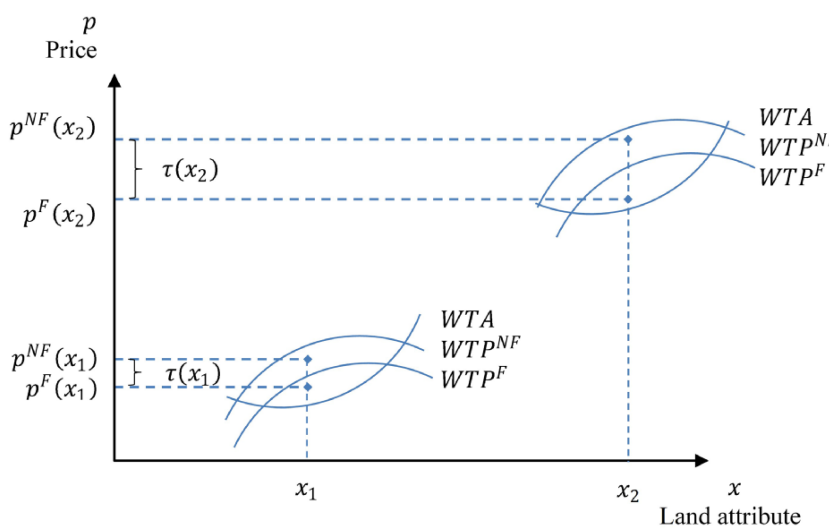

Fig. 1. The willingness to accept (*WTA*) of sellers and willingness to pay (*WTP*) of farmers (*F*) and non-farmers (*NF*) as a function of the land attribute x.
Note: p^F and p^{NF} present potential transaction prices for farmers and non-farmers, respectively. $\tau(x)$ denotes the price differential between buyer types conditional on land attributes (treatment effect).

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Systematic differences in the WTP for land may arise from the cost of financing. Financial investors are presumed to be endowed with equity capital and high liquidity, while (smaller) farms may have to borrow money to (co)finance land purchases and may thus face credit constraints. Due to capital market imperfections, the costs of credits are usually higher than opportunity costs of equity capital. Hence, the discount factor in Equation (1) will be smaller for farmers than for financial investors. A further argument provided in the literature refers to the diversification potential of land if it is added to a financial portfolio (Hüttel *et al.*, 2013). According to the Capital Asset Pricing Model, the expected return of an asset is determined by the correlation between the asset's return and the return of the market portfolio. The lower the correlation is, the smaller is the required risk premium on top of the riskless return. To use this relation as an argument for the existence of a potential wedge between the WTP of farmers and non-farmers, we must impose the assumption that non-agricultural investors have more diversified portfolios than farmers, so that the latter require higher risk premia on their farmland investments. Family farms typically re-invest their profits in land, buildings and machinery, i.e. within the agricultural sector, while at least some financial investors active in the German land market are from other sectors, so farmland investments help diversify their financial portfolios (Forstner *et al.*, 2011).

However, farmers are not generally in a disadvantaged position when competing with non-farmers for land. First, farmers as owner-operators retain the entire land rent, while landowners, who do not operate the land, must share the land rent with the tenant. Typically, lease rates are lower than land rents because the latter includes a risk premium. Second, farmers as potential buyers are often better informed about the traits of a land plot and its income potential (e.g. Hüttel *et al.*, 2013) and thus they require smaller risk premia to account for uncertainties of land returns (Seifert, Kahle and Hüttel, 2021). This, in turn, leads to a higher WTP compared with (non-local) non-agricultural investors. Third, farmers are often located closer to the offered land tracts than foreign or transregional investors. Remoteness of landowners is a disadvantage when negotiating land lease rates with tenants (Kuethe and Bigelow, 2018; Kionka *et al.*, 2022). This should lead to a lower WTP for farmland of non-local investors.

The net effect of the aforementioned determinants of the land buyers' WTP is unclear and must be measured empirically. This measurement is challenged by the fact that markups and markdowns to the WTP of farmers and non-farmers are not necessarily constant, but may rather vary with land attributes, such as parcel size, distance to the nearest highway exit or soil quality. For example, non-farmers are willing to pay higher land prices for small land plots that are used for recreational purposes ('hobby farming'), particularly in the urban fringe (Stobbe, Eagle and van Kooten, 2008). This means that they assign an intrinsic value to the ownership of land that is not captured in the present value formula (Equation 1) (Brorsen, Doye and Neal, 2015). This 'small parcel size premium', however, vanishes with increasing plot size because in contrast to professional farmers, economies of size do not play a

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role for hobby farmers. Also the price-markup of private local non-farmers who want to diversify their financial portfolios typically applies to smaller plots because of a limited amount of equity capital (Curtiss and Forstner, 2020; Curtiss *et al*., 2021).[1]

Another example is the location of the plot. Access to infrastructure, such as distance to a highway, can affect the WTP of farmers and non-farmers differently. For non-farmers, a shorter distance to a highway could translate into positive implications, primarily because such land is perceived as having a higher potential for commercial development or other non-agricultural uses (Qiu *et al*., 2015). In contrast, the presence of a highway can pose several challenges for farmers. These include potential environmental pollution (Kibblewhite, 2018) as well as disruptions in cost-efficient farming operations and constraints on the expansion of agricultural activities (Seifert, Kahle and Hüttel, 2021) since highways are more difficult to cross and are not allowed to be used by tractors. Therefore, we expect that a price markup by non-farmers will decline with increasing distance to a highway exit as this measure is used for measuring remoteness and accessibility.

A third moderating effect could be soil quality since farmers and non-farmers may have different motivations for buying farmland. Soil quality is a major factor in determining the WTP for farmland, especially for farmers who intend to use the land for agricultural production (Huang *et al*., 2006). Non-farmers, who are often interested in non-agricultural uses, may not consider soil quality as a significant factor in their valuation of land. This disparity in valuation criteria between the two groups suggests that soil quality could influence the gap in WTP between a farmer and non-farmer. It is hypothesised that the difference in WTP will vary in response to increasing soil quality. We assume a decrease in the potential price markup by non-farmers.

These examples illustrate that assessing the impact of non-farmers differs from the classical treatment effect measurement in which 'cause' and 'effect' can be clearly described. The subject in our case is not a firm, household or consumer, but a tract of land that has been sold. Treatment refers to the type of investor that bought the land. Clearly, this treatment cannot be randomly assigned. The treatment effect is understood as the price differential of a land plot that a seller could realise if the land were sold to a non-farmer instead of a farmer. Apparently, this effect results from an interplay of land traits, on the one hand, and the valuation, characteristics and capabilities of buyer types, on the other hand.

1 Plot size may not only have a moderating effect on observed price differentials via the WTP of farmers and non-farmers. Price mark-ups may also vary with plot size if land market regulations are related to this variable. For example, price caps or preferential treatment of farmers may only be effective for larger plots, while smaller plots are unregulated. We are grateful to an anonymous reviewer for emphasising this effect.

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3. Empirical framework

3.1. Definitions and assumptions

As described in Section 2, we consider the buyer type—farmer or non-farmer—as a 'treatment' of the ith plot that can be modelled as a binary treatment variable W_i, [0; 1] of plot i. $Y_i \in \mathbb{R}$ denotes the price of plot i with observable characteristics $X_i \in \mathbb{R}^p$. To estimate the treatment effect of the buyer type, we refer to the potential outcome framework of Rubin (1980) and Imbens and Rubin (2015). Let $Y_i^{(0)}$ and $Y_i^{(1)}$ denote the potential outcomes of the ith subject receiving treatment ($W_i = 1$) or no treatment ($W_i = 0$), respectively. $Y_i = Y_i(W_i)$ is the observed outcome. In this framework, the conditional average treatment effect (CATE) is defined as (Wager and Athey, 2018; Athey and Wager, 2019):

$$\tau(x) = E\left[Y_i^{(1)} - Y_i^{(0)} \mid X_i = x\right]. \tag{2}$$

To identify treatment effects from observable data, the following assumptions must be fulfilled. First, unconfoundedness (conditional independence) implies that the treatment assignment is as good as random conditionally on covariates (Rosenbaum and Rubin, 1983):

$$\left\{Y_i^{(0)}, Y_i^{(1)}\right\} \perp W_i | X_i. \tag{3}$$

Second, the overlap assumption (common support) must be fulfilled, which means that all land plots have a non-zero probability of being treated (i.e. being bought by a non-farmer) and a non-zero probability of not being treated (i.e. being bought by a farmer):

$$0 < P[W = 1 | X = x] < 1. \tag{4}$$

A further assumption that must be satisfied is the Stable Unit Value Assumption (SUTVA), which states that the potential outcome of a land transaction is not affected by the treatment assignment of other land transactions, but is only affected by its own assignment (see Rubin, 1980).

Under unconfoundedness, the CATE from Equation (2) can be expressed as follows (Wager and Athey, 2018):

$$\tau(x) = E\left(Y_i\left(\frac{W_i}{e(x)} - \frac{1 - W_i}{1 - e(x)}\right) | X_i = x\right), \tag{5}$$

where $e(x)$ denotes the propensity score and $e(x) = P(W_i = 1 \mid X_i = x)$, i.e. the probability of receiving treatment for given covariates x. To estimate the CATEs $\tau(x)$, which depend on the propensity score $e(x)$, we apply causal forests following the procedure proposed by Wager and Athey (2018) and Athey, Tibshirani and Wager (2019). This choice is motivated by the fact that causal forests are computationally efficient and robust against confounding. In the next section, we briefly outline the estimation procedure.

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3.2. Estimation of heterogeneous treatment effects with causal forests

A major challenge in estimating treatment effects and potential treatment effect heterogeneity is modelling the unknown functional form $E\left[Y^{(1)} - Y^{(0)}|X\right]$ (Langen and Huber, 2023). Unlike parametric estimation methods, which require manually specifying the functional form of the relationships between variables, the causal forest approach developed by Wager and Athey (2018) and Athey, Tibshirani and Wager (2019) provides a data-driven, non-parametric method to estimate the underlying functional form directly from the data, without needing to explicitly define the functional relations by hand. Causal forests make use of (generalised) random forests in the context of causal effect estimation. As with classical estimators, e.g. K-nearest neighbours matching, they aim at finding neighbourhoods in the covariate space. Rather than defining a neighbourhood $N(x)$ by means of a distance measure, neighbours of a test point x are those observations in the training set that fall into the same leaf L of a causal tree x. As a result, pairs of treated/non-treated individuals within a leaf can be considered as an outcome of a randomised experiment for which the treatment effect can be estimated as (Jacob, 2021)[2]:

$$\hat{\tau}(x) = \frac{\sum_{i:\, X_i \in N(x)} (Y_i - \hat{\mu}(X_i))(W_i - \hat{e}(X_i))}{\sum_{i:\, X_i \in N(x)} (W_i - \hat{e}(X_i))^2}, \tag{6}$$

where $\mu(x) = E(Y_i|X_i = x)$, i.e. the expected land price for a given set of land attributes. To implement Equation (6), the propensity score $e(x)$ and the conditional mean function $\mu(x)$ (the so-called 'plug-in-parameters') have to be estimated, which can be attained by any predictive machine learning algorithm (Langen and Huber, 2023).

The challenge is to find neighbourhoods for which the CATE can be assumed to be constant. This objective is inherent to the construction of the causal tree. Leaves of a causal tree are constructed by recursively splitting the covariate set into subgroups based on heterogeneity in treatment effects. The splitting rule rests on an adapted criterion function that rewards a split that increases the variance of the treatment effects across leaves and penalises a split that increases within-leaf variance (Athey and Imbens, 2016; Baiardi and Naghi, 2024).

Eventually, the CATE estimate is derived from a forest, i.e. multiple causal trees, each of which is grown on a subset of the entire sample. The outcome of a random forest can be considered as an average of estimates over B individual trees. Athey, Tibshirani and Wager (2019) provide an alternative motivation of a random forest as a weighting function:

$$\hat{\tau}(x) = \frac{\sum_{i=1}^{N} \alpha_i(x)\left(Y_i - \hat{\mu}^{-i}(X_i)\right)\left(W_i - \hat{e}^{-i}(X_i)\right)}{\sum_{i=1}^{N} \alpha_i(x)(W_i - \hat{e}^{-i}(X_i))}. \tag{7}$$

2 Note that Equation (6) resembles an R-learner suggested by Robinson (1988).

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The superscripts $-i$ in Equation (7) indicate cross-validation (also referred to as 'out-of-bag prediction'), which means that the estimate is computed without observation i (Athey and Wager, 2019). The weights α_i are given by:

$$\alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B}\frac{1\{X_i \in L_b(x)\}}{|L_b(x)|}. \tag{8}$$

The numerator in Equation (8) counts how often an observation i falls into the same leaf as the test point x. The denominator is the total number of observations in leaf L_b.

The estimation is conducted in two steps. In the first step, two separate regression forests are grown to estimate the plug-in-parameters $\hat{e}$ and $\hat{\mu}$, from which residuals $Y_i - \hat{\mu}(X_i)$ and $W_i - \hat{e}(X_i)$ are calculated. In the second step, a causal forest is developed, where at each potential split, the following regression is estimated (Nie and Wager, 2021):

$$Y_i - \hat{\mu}(X_i) = (W_i - \hat{e}(X_i))\,\hat{\tau}(X_i) + \varepsilon_i, \tag{9}$$

where ε_i denotes an error term. By comparing the estimated CATEs $\hat{\tau}$ for all potential splits, the tree structure that maximises the heterogeneity of treatment effects can be determined. Chernozhukov *et al.* (2017) show that this two-step procedure using a 'residuals-on-residuals' approach is semiparametrically efficient under unconfoundedness. The intuitive reasons are that Equation (9) is much simpler compared with the propensity score function $e(x)$ and conditional mean function $\mu(x)$, as well as that estimates for $\tau(x)$ converge faster than those for $e(x)$ and $\mu(x)$ (Nie and Wager, 2021). The calculations are carried out in *R* by means of the causal forest function in the *grf* package provided by Tibshirani *et al.* (2022).

3.3. Estimation of ATEs and group average treatment effects

The procedure described above results in a causal forest from which individualised treatment effects can be derived for any test point x in the covariate space. Individualised average treatment effects represent the finest aggregation level of analysis (Knaus, Lechner and Strittmatter, 2021). It is possible, however, to calculate more aggregated treatment effects, particularly ATEs and group average treatment effects (GATEs). While ATEs average individual effects over the entire population, GATEs relate to subpopulations and can be used to illustrate heterogeneity of treatment effects (Knaus, Lechner and Strittmatter, 2021). To estimate ATEs and GATEs, we follow Athey and Wager (2019) and use augmented inverse propensity weighting (AIPW) through the following equation:

$$\tau = \frac{1}{N}\left(\sum_{i=1}^{N}\hat{\tau}^{-i}(X_i) + \frac{\left(W_i - \hat{e}^{-i}(X_i)\right)}{\hat{e}^{-i}(X_i)(1-\hat{e}^{-i}(X_i))}\left(\left(Y_i - \hat{\mu}^{-i}(X_i)\right)\right.\right.$$
$$\left.\left. - \left(W_i - \hat{e}^{-i}(X_i)\right)\hat{\tau}^{-i}(X_i)\right)\right). \tag{10}$$

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The estimates $\hat{\tau}^{-i}$, $\hat{e}^{-i}$ and $\hat{\mu}^{-i}$ in Equation (10) are obtained from the estimated causal forest. The basic idea of AIPW as a doubly robust method is to utilise not only the estimated propensity score but also estimates of the conditional mean function to make the classical inverse propensity weighting more robust against very low or high values of propensity scores. The AIPW estimator is asymptotically normally distributed and consistent if either the assignment function $e(x)$ or the outcome function $\mu(x)$ is correctly specified (Glynn and Quinn, 2010). Both ATE and GATE are calculated with the function 'average_treatment_effect' from the *grf* package with standard errors clustered at the transaction level.

In summary, the causal forest procedure described above shows several advantages. It goes beyond the calculation of ATEs by accounting for potential heterogeneity and a more fine-grained analysis of treatment effects. It provides a data-driven method to estimate the underlying functional form without having to specify the functional relations as is the case in parametric estimation methods. Large covariate sets can be easily accommodated. Causal forests show desirable statistical properties. They provide consistent estimates of the true treatment effect (Wager and Athey, 2018) and as ensemble methods they combine low bias with low variance (Athey, Tibshirani and Wager, 2019). Moreover, Athey and Wager (2019) show that causal forests are robust to confounding even when not all confounders are included in the analysis.

4. Study region and data

Germany is one of the most important countries in the European Union in terms of agriculture. In the European Union, Germany's agricultural industry output ranked second in 2019 at 58.2 billion EUR, and pork and dairy production were higher than in any other EU Member State (Eurostat, 2021). Although Germany has a high population density, half of its land is used for farming (BMEL, 2020). While the average price for 1 ha of agricultural land is 26,777 EUR (2020), there is large regional heterogeneity at the federal state level, ranging from 10,687 EUR (Saarland) to 63,986 EUR (Bavaria) (Destatis, 2021). Our chosen study area is the federal state of Lower Saxony, which has an average value of farmland of 40,916 EUR/ha and is located in northwest Germany. The highest number of farmland transactions in 2020, 4,455 or 13 per cent of all farmland transactions in Germany, took place in Lower Saxony. Lower Saxony also has the highest agricultural output (23 per cent) of any German federal state (Statistikportal, 2022) and has large regional heterogeneity, which renders it an interesting study region for many studies (e.g. Yang, Odening and Ritter, 2019; Plogmann *et al.*, 2022; Schaak and Musshoff, 2022).

The regulation of the land market in Germany falls within the jurisdiction of federal states. The legal framework for federal states to pass specific laws is provided by the Land Transfer Act (*Grundstückverkehrsgesetz*). In 2022, the state of Lower Saxony introduced a comprehensive law regulating the sales of farmland (Niedersachsen, 2022). However, our study period 2005–2019 falls outside the scope of the current law. According to state law in Lower Saxony

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before 2022, authorisation is required for the sale of farmland exceeding 1 ha (Niedersachsen, 1970). This authorisation may be denied only if the sale would result in an unhealthy distribution of land or if it would economically reduce or divide the parcel or most of the parcels that are spatially or economically connected. Furthermore, authorisation may also be denied if the equivalent value is vastly disproportionate to the sales price of the plot, but this only applies if the land continues to be used for agricultural purposes. This significant discrepancy is presumed when the intended sales price varies by approximately 50 per cent from the average local market price (Wissenschaftliche Dienste des Deutschen Bundestages, 2023). This regulation does not differentiate between farmers and non-farmers, and the proportionality between price and land value must be met by both types of investors. Regarding non-farmers, no explicit regulations exist during our study period in either state or federal law (Niedersachsen, 1970; Bundesrepublik Deutschland, 2008). Nevertheless, licensing authorities may perceive non-farmer status as having an adverse effect on the agricultural structure, leading to a potential basis for rejection.

In all these instances, non-profit settlement companies for the development of rural areas (*Landgesellschaften*) are tasked with reviewing pertinent cases and, if required, declining the transaction and exercising the right of first refusal. However, this right of first refusal can only be exercised if there is an alternative farmer willing to purchase the land on the same terms as the non-farmer. There is no explicit data for such matters in Lower Saxony; however, at the national scale, the settlement companies for the development of rural areas review an average of approximately 2 per cent of land sales in Germany, with only about a quarter of these cases exercising the right of first refusal. The number of registered cases per year varies considerably from state to state. In Lower Saxony, there were between 20 and 60 cases per year from 2012 to 2019 (BLG, 2023).

The main data set used in this study contains farmland sales transactions in Lower Saxony between 2005 and 2019 and was collected by the Committee of Land Valuation Experts in Lower Saxony (*Oberer Gutachterausschuss für Grundstückswerte in Niedersachsen*). The most important information for our application is the indicator of the buyer type since this is the basis of our definition of treatment. This information is not mandatory but is collected in many cases by the Committee of Land Valuation Experts through a questionnaire when a transaction occurs. Thus, we rely on the classification by experts. From the original data set containing 106,242 sales transactions of arable land and grassland, we first remove all observations with missing soil quality information (13,829 observations) or missing information on the buyer and seller type (53,095 observations). According to the Committee of Land Valuation Experts in Lower Saxony, the observations with buyer information can be considered as representative for the full data set (Oberer Gutachterausschuss für Grundstückswerte in Niedersachsen, 2019). Our own comparison of the groups with and

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without buyer and seller information does not reveal substantial differences in the means of key variables either.[3]

The existing buyer groups consist of farmers, private entities, public entities and non-farmers. We further exclude the group of private entities as buyers (e.g. sugar refineries, limited partnerships, registered cooperatives or associations) due to the inability to distinguish between farmer and non-farmer buyers (1,042 observations), as well as the group of public entities (such as the federal state of Lower Saxony, counties, churches, public banks or foundations) since they pursue public interests and hence differ strongly in the motives and behaviour compared with private non-farmers (1,745 observations).

For the remaining 36,531 observations, the buyer categories are delineated as 'farmer' and 'non-farmer'. The 'farmer' category, which represents the control group, includes buyers who are either individuals or sole proprietors, without distinction as to their previous tenant status. This group accounts for 56.9 per cent of the transactions in the data set. Conversely, the 'non-farmer' category, designated as the treatment group, comprises the remaining 43.1 per cent of the observations classified as non-farmers.

To achieve the final data set, we conduct further data cleaning. We first remove two observations with implausible soil quality information (below 1 and above 120), as well as 439 observations with a parcel size below 100 sqm and 18 observations with a price below 100 EUR/ha. The round values of 100 sqm or 100 EUR/ha represent a compromise, balancing the exclusion of observations with exceptionally low values and low relevance, with the goal of retaining as many observations as possible. Moreover, we drop 975 observations labelled as not suitable for analysis by the Committee of Land Valuation Experts and 65 observations with unknown special circumstances regarding the transaction. Lastly, we exclude 16 observations labelled as 'exercised right of first refusal' (0.05 per cent), leading to a final data set of 35,016 observations. For each transaction, plot-specific information (parcel size, soil quality, type of land and location at the district [*Gemarkung* level]) and details about the transaction (price, date of transaction, information on the buyer/seller and further transaction information) are available. These variables will be explained in the next subsection based on the detailed data description by Landesamt für Geoinformation und Landvermessung Niedersachsen (2019).[4]

The average sales price for farmer buyers is 18,703 EUR/ha (Table 1), which differs from the average price for non-farmer buyers, 17,776 EUR/ha. This

3 Observations lacking information about the buyer or seller tend to exhibit a slightly higher average price compared to those with complete information (22,895 EUR/ha vs. 19,309 EUR/ha), accompanied by a slightly smaller parcel size (2.04 ha vs. 2.21 ha) and lower soil quality (42.48 vs. 44.45 index points). Furthermore, these transactions typically occur slightly later in time, with an average transaction year of 2012.1 for the data set with missing information, compared to 2010.5 for the reduced data set with complete information.

4 The data set may contain multiple purchases by the same buyer. This should not pose a problem to the estimation of treatment effects for land plots as long as his/her preference and thus WTP is representative for the group that person represents. A potential issue could arise if the buyer's WTP is atypical and given greater weight through multiple purchases. Unfortunately, we are unable to identify multiple purchases due to the absence of buyer identity information in our data set.

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Table 1. Descriptive statistics by buyer type

	Farmer buyer		Non-farmer buyer	
	$N = 20{,}375$		$N = 14{,}641$	
Variable	Mean	SD	Mean	SD
Price (EUR/ha)	18,702.696	14,361.483	17,775.990	18,084.044
Plot-specific				
Parcel size (ha)	2.658	3.645	1.586	3.286
Soil quality (index)	43.902	17.445	44.836	16.659
Grassland (D)	0.387	–	0.421	–
Distance to highway exit (km)	13.357	9.642	13.426	9.740
Transaction-specific				
Year (1 = 2005)	6.821	3.915	5.946	3.903
Seller type				
Farmer (D, reference)	0.345	–	0.182	–
Non-farmer (D)	0.620	–	0.805	–
Public entity (D)	0.019	–	0.007	–
Private entity (D)	0.016	–	0.006	–
Special circumstances				
None (D, reference)	0.770	–	0.822	–
Consolidation purchase (D)	0.116	–	0.122	–
Family relation (D)	0.023	–	0.039	–
Tenant buyer (D)	0.082	–	0.005	–
Forced sale (D)	0.004	–	0.005	–
Collector's price (D)	0.001	–	0.002	–
Employment relation (D)	0.001	–	0.003	–
Alternative land (D)	0.002	–	0.004	–
County-specific				
Hog density (per 100 ha AA)	244.248	421.538	256.631	383.366
Cattle density (per 100 ha AA)	98.989	66.112	75.792	57.577
Population density (1/sqkm)	179.409	148.493	186.071	145.760
Population's average age	43.865	1.893	43.676	1.890
Net reproduction rate	–0.003	0.002	–0.003	0.002
Migration rate	0.003	0.005	0.002	0.005
Agricultural area (share)	0.635	0.114	0.599	0.119
Leisure area (share)	0.011	0.008	0.011	0.007
Industrial area (share)	0.012	0.008	0.012	0.009
Street area (share)	0.037	0.012	0.040	0.012
Residential area (share)	0.043	0.017	0.042	0.017
Planned biomass (kW/ha)	0.020	0.034	0.022	0.033
Planned wind energy (kW/ha)	0.028	0.151	0.025	0.158

Note: D denotes a dummy variable; AA stands for agricultural area.

means that farmers, on average, pay 927 EUR/ha more for land than non-farmers. However, this difference cannot be interpreted as the ATE since we cannot assume unconfoundedness without further corrections.

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4.1. Covariates

The explanatory variables used in our analysis can be categorised into plot-specific, transaction-specific and county-specific variables. Plot-specific information comprises the size of the sold plot (parcel size), its soil quality, the type of land and its accessibility. The average parcel size for farmer buyers is 2.658 ha, which is larger than that for non-farmer buyers (1.586 ha). Soil quality is measured by an index for the productivity of arable land (*Ackerzahl*) or grassland (*Grünlandzahl*), which ranges from 4 (very low productivity) to 103 (very high productivity) in our data set.[5] The average soil quality also differs between the buyer groups (43.9 for farmers vs. 44.8 for non-farmers). The type of land, i.e. arable land or grassland, is a dummy variable (1 = grassland). Non-farmers purchase grassland slightly more often compared to farmers (42 per cent vs. 39 per cent, respectively). As a measure for the accessibility of the plot, we use the distance to the nearest highway exit, which is calculated as the distance as the crow flies from the centre of the local administrative unit where the transaction took place.[6] The average distance is around 13 km for both buyer types.

Besides the sales price, further transaction-specific information is available. We consider the year in which the transaction took place as a linear variable (1 corresponds to 2005). Another variable provided by the Committee of Land Valuation Experts is the seller type, which has the same categories as the buyer type. Keeping all four categories, we obtain that most transactions are sold by non-farmers (62 per cent for farmer buyers and 81 per cent for non-farmer buyers), followed by farmers as sellers (34 per cent and 18 per cent), public entities (1.9 per cent and 0.7 per cent) and private entities (1.6 per cent and 0.6 per cent).

The Committee of Land Valuation Experts also provides information about special circumstances with respect to the transaction that indicates whether the sales price might be influenced by unusual or personal circumstances. These special circumstances are captured as dummy variables. Around 79 per cent of the transactions have no special circumstances, 12 per cent are consolidation purchases (e.g. for rounding off a property), and 3 per cent are transactions between family members (more frequent among non-farmer buyers). In 8 per cent of the transactions with farmer buyers, the purchase price could have been influenced by the fact that the buyer was already the tenant of the land sold—a circumstance that is very rare for non-farmer buyers (0.5 per cent). The additional causes (forced sale, collector's price, employment relation and alternative land[7]) each occur in less than 1 per cent of the transactions.

5 If soil quality values for both arable land and grassland are provided for one transaction, we take the value according to the classification of the plot as arable land or grassland.

6 The locations of highway exits are retrieved from *Niedersächsische Landesbehörde für Straßenbau und Verkehr* in 2022.

7 Forced sales encompass emergency sales, as well as sales resulting from insolvency proceedings and bankruptcy. Collector's price refers to a value that reflects the collector's interest rather than the market value. Employment relation implies that there is an employment connection

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To capture regional heterogeneity and to reduce the risk of omitted variable bias, we take the locations of the transactions and merge them with structural data measured at the county level derived from the German agricultural census for lower Saxony (Niedersächsisches Ministerium für Ernährung, Landwirtschaft und Verbraucherschutz, 2020). Data are collected for the years 1999, 2007, 2010, 2016 and 2020 and are linearly interpolated for years between. Regarding the regional agricultural structure, we include the density of hog farms and cattle farms, measured per 100 ha of agricultural area at the county level. The general structure of the counties, which can also influence pressure on agricultural land, is considered through population density (1/sq km), the population's average age, the annual net reproduction rate, the yearly migration rate, as well as the shares of the total land of the county that are used for agriculture (62 per cent, on average), leisure (1 per cent), industry (1 per cent), streets (4 per cent) and housing (4 per cent). Since the use of renewable energies in a county provides a measure for an alternative use of farmland or an additional source of income (Myrna, Odening and Ritter, 2019), we include planned biomass and wind energy power, retrospectively. These are derived as the additional power from these sources installed in the county during the 4 years following a transaction (comparable to Ritter *et al*., 2015). To address variations in county sizes, we normalise these values by dividing them by the county size, resulting in the unit kW/ha.

5. Identifying assumptions and model calibration

5.1. Overlap assumption

The estimated propensity scores in Figure 2 document that the overlap assumption is fulfilled for the treatment groups since all $\hat{e}(x)$ are above 0 and below 1 (Equation 4) (Imbens and Rubin, 2015: 314). However, in case of a misclassification, $\hat{e}(x)$ close to 0 or 1 would imply a proportionally large weight for these observations in the AIPW estimation (Equation 10). To overcome this issue, we filter for $\hat{e}(x)$ between 0.1 and 0.9 when estimating ATE and GATE as proposed by Crump *et al*. (2009). The resulting data set includes 31,466 observations with 57.2 per cent in the control group (farmer buyers) and 42.8 per cent in the treatment group (non-farmer buyers). The specific parameters for the causal forest are depicted in Supplementary Table A1.

5.2. Unconfoundedness assumption

As described in Section 3, the identification of GATEs and CATEs is based on unconfoundedness. The unconfoundedness assumption is satisfied if purchases by non-farmers exhibit a quasi-random distribution among observations with identical values in X. To reduce selection bias through omitted confounders, we account for numerous variables as delineated in the preceding section,

between the buyer and seller. The circumstance of alternative land indicates land acquisitions due to compensatory measures.

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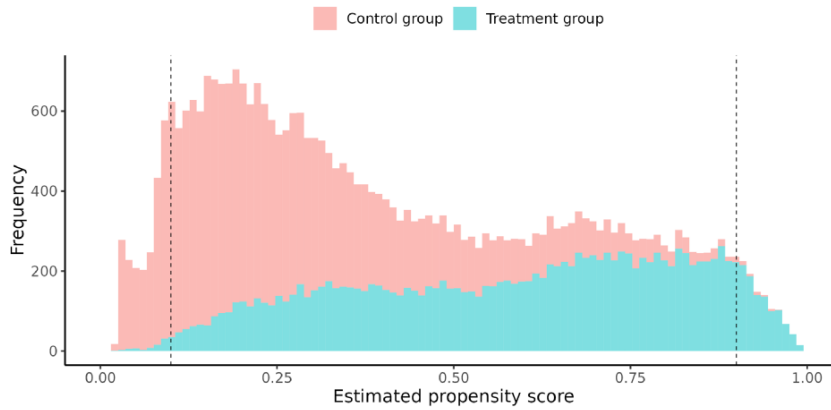

Fig. 2. Estimated propensity scores with causal forest for control group and treatment group (stacked); vertical lines indicate 0.1 and 0.9, the thresholds for filtering.

including a time trend and regional variables for capturing potential unobserved influences. Furthermore, the use of causal forests not only facilitates highly non-linear and complex interactions among covariates but also enables the method to potentially account for unobserved confounders that could be indirectly represented through observed confounders. To assess violations of unconfoundedness, we estimate pseudo-treatments effects (Rosenbaum, 1987; Imbens and Rubin, 2015). Instead of analysing the effect on the actual treatment, this approach considers the outcomes of pseudo-treatments, which are derived by dividing the control group into subgroups. This allows for the assessment of whether the control group itself is unconfounded and therefore indirectly assesses if unconfoundedness is plausible for the complete data set. The general theoretical background to this approach is described in Imbens and Rubin (2015: 485–486). Details about the implementation of the pseudo-treatment approach in our context are provided in Section A.1 in the appendix. In brief, we divide the control group of farmer buyers into former tenants and non-tenants and consider the tenants as the pseudo-treatment group. Then, we fit a separate causal forest for this subgroup and estimate the ATE. Since this estimated average effect of the pseudo-treatment is not significantly different from zero, the assumption of unconfoundedness is supported for the whole data set. As a further assessment of unconfoundedness, we check the robustness of the covariate selection and the outcome variables (see below), as proposed by Imbens and Rubin (2015: 479–495).

A challenge to our analysis is the potential violation of unconfoundedness due to existing land market regulations in our study region. Most importantly, they include an approval procedure for transactions involving land parcels larger than 1 ha that may lead to a denial of sales to non-farmers (cf. Section 4). As a result, smaller plots might have a higher probability of being treated

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(i.e. being bought by a non-farmer). This self-selection is partly controlled for by the propensity score, which depends on plot size (among other covariates). Nevertheless, there is a potential bias that we cannot control for, namely endogenous entry. Potential non-farmers might be deterred by restrictive regulations and decide not to participate in the land market. Moreover, even participating non-farmer buyers may not reveal their true WTP because they anticipate that high offers might be considered as speculative and incommensurate with the 'true' value and thus are not approved. Such a capping of offers by non-farmers would affect the interpretation of the conditional treatment effect of plot size. A declining treatment effect would then not only reflect a converging WTP of farmers and non-farmers but also a strategic response of non-farmers to land market regulations. A proper identification of land market regulations' influence on the treatment effect is difficult. The endogenous entry problem cannot be analysed with observational data (transaction data) and would require a survey including not only actual but also potential land buyers.

Though we cannot rule out the aforementioned effects of land market regulations on conditional treatment effects, we are confident that they are only modest. Land market regulations in Germany are considered as liberal compared with other states, such as France or Poland (Vranken *et al.*, 2021). The fact that only 20–60 interventions by the authorities have been reported per year in Lower Saxony supports this presumption for our study region. Moreover, some criteria for a denial of land transactions, such as unfair pricing, apply to farmers as well as to non-farmers and should not affect the price differential between both investor types.

5.3. Stable Unit Treatment Value Assumption

In the realm of agricultural land markets, potential violations of the SUTVA could arise due to (unmodelled) spatial and temporal spillovers. To mitigate potential violations of SUTVA due to spatial and temporal spillovers, we incorporate regional and temporal variables as covariates in our causal forest model. By doing so, we aim to capture any potential spillover effects that might be associated with the treatment assignment of other units in the same region or time (Gelman and Hill, 2006). It is important to note that not every spillover effect implies a violation of the SUTVA. For example, the presence of a dominant buyer might affect the bidding behaviour of other market participants and thus price formation on a local land market (Balmann *et al.*, 2021). However, a violation of the SUTVA only occurs if the response of market participants depends on whether the dominant buyer is a farmer or non-farmer. We cannot rule this out completely, but to our best knowledge, the empirical literature does not provide evidence that farmers and non-farmers exercise market power differently. Nevertheless, a violation of the SUTVA could arise from the presence of non-farmers in a local market, as they not only affect the prices of the plots they purchase but also the bidding behaviour of farmers. Farmers might bid more aggressively to prevail against competing financial investors. Fortunately, Credit and Lehnert (2023) demonstrate by means of simulation

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studies that causal forests are robust against violations of the SUTVA caused by spatial spillovers.

5.4. Model calibration and robustness checks

Before interpreting causal forest algorithm results, it is essential to verify the model's calibration accuracy. For this purpose, we performed a weighted linear regression of the model's predictions. An ideally calibrated model shows a linear relationship between observed outcomes and predictions, with regression coefficients near unity. The function also allows for heterogeneity assessment by modelling differences between individual and mean predictions, i.e. differential forest predictions. Significant deviation of this regression coefficient from zero indicates treatment effect heterogeneity across observations. This procedure is based on the work of Chernozhukov *et al*., (2017). An application to observational data can be found in Athey and Wager (2019). However, the results should be interpreted cautiously and do not allow inference, but instead provide a good starting point for further analysis of heterogeneity in treatment effects. The calibration outcomes of the causal forest exhibit values close to one and show significant results for differential forest prediction, allowing us to reject the null hypothesis of no heterogeneity. Detailed results are shown in Supplementary Table A2 in the appendix.

We check the robustness of the causal forests against misspecifications by placebo test, a variety of leave out assessments, and the introduction of simulated confounders with diverse correlation structures as proposed in Imbens and Rubin (2015). Further, by simulating a target common cause correlated with both treatment and outcome, we examine the model's sensitivity to potential latent factors not encapsulated in our covariate space. Details and results of these robustness checks are documented in Appendix A.3. The findings indicate that the estimated models are robust regarding the tested modifications.

We also stress the reliability of the causal forest itself. Machine learning models offer a new approach to estimation that traditional econometric methods can only partially address. To see, however, if the causal forest approach is consistent in the parts where it can be compared to traditional econometric approaches, we compare it with several econometric benchmark models. Supplementary Table A3 shows that the treatment effects estimated with the benchmark models are slightly lower than those obtained with the causal forest method, but the difference is not substantial when looking at the overlapping confidence intervals. All methods result in a change in sign compared to the mean value comparison. The benchmark models exhibit lower precision (larger confidence intervals) than the causal forest as well lower R^2 and pseudo-R^2 (a detailed description can be found in Appendix A.4).

6. Results and discussion

6.1. Average and conditional treatment effects

Using a causal forest to control for amenities, we find that non-farmer buyers pay a higher price for farmland on average than farmers. This is reflected in

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the ATE, which differs significantly from the difference in means approach (Section 4). It amounts to +1,998 EUR/ha, translating to a price markup of +10.9 per cent over the average sales price of 18,335 EUR/ha. The critical interval ($\alpha = 0.05$) is [+1,723 EUR/ha; +2,274 EUR/ha] ([+9.4 per cent; +12.4 per cent]), confirming that the ATE is statistically significant and of an opposite sign to the simple sample mean difference.[8]

Moreover, we observe considerable heterogeneity in the ATE. The estimated CATEs $\hat{\tau}(x_i)$ range from –11,257 EUR/ha to +64,4370 EUR/ha with 95 per cent between –1,547 EUR/ha and +9,596 EUR/ha, as illustrated in Figure 3. This suggests that the higher valuation by non-farmers is not uniform across all observations, indicating a need for further analysis to understand the underlying factors driving this heterogeneity.

6.2. Group average treatment effects

Next, we investigate whether soil quality, distance to highway exit and plot size have moderating effects on the treatment effect. Our results show that parcel size and distance to highway exit have a clear moderating effect, as shown by the GATEs calculated after dividing our data set into five groups for each variable based on the quintiles of the filtered data set (see descriptive statistics in Supplementary Table A4).

The GATEs for parcel size indicate a moderating effect on the price difference between farmers and non-farmers. Non-farmer buyers pay significantly more for smaller parcels, with an average price markup of 22.7 per cent/ha in the smallest parcel size group compared to 3.7 per cent in the largest parcel size group, as shown in Figure 4a. The result of a Wald test supports the finding of heterogeneity, as it rejects the null hypothesis of no significant difference between the parcel size groups ($p = 4.64 \cdot 10^{-14}$). This suggests that parcel size not only influences the general price of land, as noted in prior research (see e.g. Brorsen, Doye and Neal, 2015; Ritter *et al.*, 2020; Schaak, Meissner and Musshoff, 2023), but also plays a role in the price markup paid by non-farmers.

As the presence of regulation is triggered by the parcel size (>1 ha), the differentiation between the impact of parcel size and land market regulation is challenging. On the one hand, the heterogeneity in price markups could be driven by different valuations of parcel size between farmers and non-farmers (see Section 2 for a detailed discussion). On the other hand, non-farmers could perceive easier access to smaller parcels due to the lack of regulation and thus pay more than farmers. It is important to acknowledge the possible presence of a bias in our results, as discussed in Section 2. The regulation could have an unmeasured impact that is translated through parcel size into our findings. This bias could arise from non-farmers anticipating that offering too high a selling price for a regulated parcel may result in the transaction being rejected, potentially leading to an underestimation of the price markup paid by non-farmers. However, while we cannot fully control for this potential bias, the

8 The treatment effect estimated by benchmark models leads to similar results. See Appendix A.4 for a detailed presentation of the benchmark model results.

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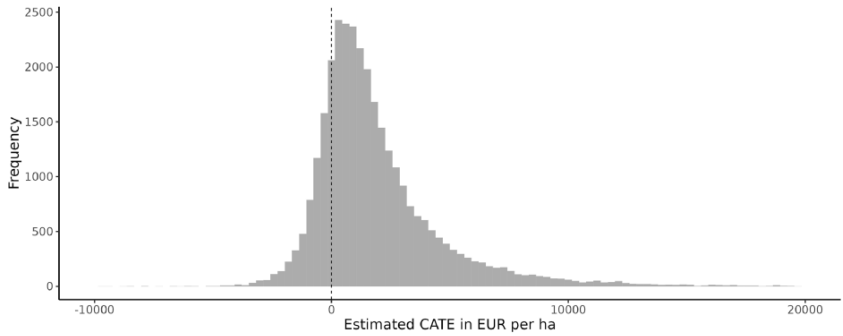

Fig. 3. Histogram of estimated CATEs in EUR/ha between 10,000 and 20,000.

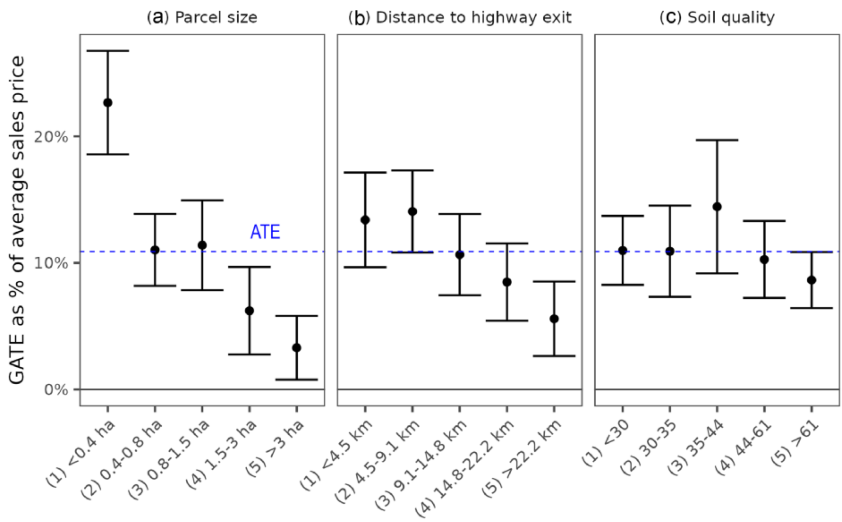

Fig. 4. Estimated GATE as % of average sales price and 95 per cent confidence intervals for (a) parcel size, (b) distance to highway exit and (c) soil quality.

fact that 30.3 per cent of the regulated plots in our data are still purchased by non-farmers indicates that this market segment remains open and accessible to non-farmers, suggesting that the impact of this bias may be limited.

While we observe the highest price markup of 22.7 per cent for the smallest parcel size group, this group represents only 1.7 per cent of the total area sold (see Supplementary Table A4). The largest parcel size group, with a GATE of 3.2 per cent, represents 61.5 per cent of the total area sold. From an agricultural point of view, larger parcels are particularly interesting because increasing parcel size leads to lower costs (Heinrichs *et al*., 2021). Therefore, the impact on the land market and agricultural relevance must also be considered from

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this additional perspective, with the most attractive parcels in terms of size having the lowest markups.

Figure 4b shows that proximity to a highway exit has a significant impact on price differences between farmers and non-farmers. This is supported by the result of a Wald test (rejecting the null hypothesis of no difference between groups, $p = 8.05 \cdot 10^{-4}$). The GATEs of the two groups closest to a highway exit (<9.1 km) are significantly higher, with 13.4 per cent and 14.1 per cent, respectively. As the distance from a highway exit increases beyond 9.1 km, the GATEs decrease. Even for parcels furthest from an exit (>22.2 km), the markup is still statistically significant, but declines to 5.6 per cent.

These results are consistent with our hypothesis that non-farmers value accessibility more than farmers. This could be due to a potentially higher likelihood of conversion of land to non-agricultural use, which is reflected in the speculative component of non-farmers' WTP. This finding sheds light on the general positive effect of accessibility or urban proximity on farmland prices that has been reported in the literature (e.g. Livanis *et al.*, 2006; Sklenicka *et al.*, 2013; Seifert, Kahle and Hüttel, 2021) and relates it to the WTP of different buyer groups. Additionally, it also supports the finding of Vyn and Shang (2021) who also find a positive impact of accessibility on price differences between farmers and non-farmers.

The GATEs for different soil quality groups (Figure 4c) show that the treatment effect is similar across different soil quality groups, ranging from 8.6 per cent to 14.4 per cent with strongly overlapping confidence intervals. A Wald test cannot reject the null hypothesis of no differences ($p = 0.30$).

This finding contradicts our initial hypothesis that soil quality is not an essential pricing attribute for non-farmers and hence would moderate the treatment effect in a significant way. The homogeneity of the treatment effect may be explained by the fact that non-farmers regularly rent out their land to farmers, so that eventually the productivity-enhancing effect of soil quality matters for all buyer types.[9]

6.3. Shapley Additive Explanation Values

While GATEs disaggregate overall treatment effects, they do not display the immediate impact of the grouping variable. For a more fine-grained analysis of the covariates' contribution to the treatment effect, we calculate Shapley values for the CATE. Originating from cooperative game theory, Shapley values provide a quantifiable metric for fairly attributing the contribution of individual covariates towards the variation in predicted outcomes (Shapley, 1953; Molnar, 2022). In this setting, each covariate in the causal forest—such as soil quality, parcel size and distance to highway exit—assumes the role of a 'player' in the cooperative game. The 'payoff' is quantified as the discrepancy between the individual prediction for a specific data point and the average of

9 We also calculated GATEs in relation to the overall average price instead of the group average price. These GATEs lie between 10.0 per cent and 13.4 per cent, but also do not significantly differ from each other, so the conclusion remains the same.

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the CATE (Blöbaum *et al.*, 2022; Stetter, Mennig and Sauer, 2022). Shapley values facilitate the equitable distribution of this 'payoff' among the covariates, effectively quantifying the marginal contribution of each covariate to the predicted outcome. This elucidates the interplay among different covariates, while concurrently accounting for potential interactions and redundancies in the data. Here we use the implementation of SHapley Additive exPlanation (SHAP) values (Lundberg and Lee, 2017) and apply a kernel smoother to aggregate individual effects.

The SHAP values for parcel size in Figure 5a support our hypothesis that parcel size itself has a negative impact on non-farmers' price markup and confirm the findings of the GATEs. The values range from –5,743.48 to +22,820.26 EUR/ha, implying considerable variation in the effect of parcel size across different predictions. Positive SHAP values dominate in smaller parcels (first two quintiles) with an average of 854.14 EUR/ha. Conversely, larger parcels (from the third quintile onwards) show a trend towards negative effects on the CATE, with the largest quintile averaging –568.67 EUR/ha.

Distance to the nearest highway exit has a substantial effect on the price premium for non-farmers, as shown in Figure 5b. SHAP values initially decrease for parcels closer to a highway exit, then increase from 30 to 50 km, peaking at 45 to 50 km, before decreasing again. This pattern is consistent with the results of the GATEs. The average SHAP values range from 500.65 EUR/ha in the first quintile to –329.09 EUR/ha in the last quintile, highlighting the negative effect of increasing distance.

Again, soil quality has no noticeable effect on the price premium paid by non-farmers, as can be seen from the lack of variability in the SHAP values. The SHAP analysis indicates a modest impact of soil quality. The lowest quintile of soil quality has an average SHAP value of –35.42 EUR/ha, while the highest quintile has a value of –15.47 EUR/ha. This suggests that non-farmers do take soil quality into account, although less so than plot size and proximity to highways. This may be because higher soil quality is attractive to both non-farmers and farmers and therefore does not significantly drive differences in WTP.

6.4. Other moderating variables for land price markups of non-farmers

Finally, we explore whether other variables in our covariate set (apart from parcel size, distance to the nearest highway exit and soil quality) may influence the CATE. To this end, we utilise the best linear projection method proposed by Semenova and Chernozhukov (2021), which provides a doubly robust fit to the linear model $\tau(X_i) = \beta_0 + X_i\beta$.

Table 2 depicts the results of the best linear projection. The results for parcel size, distance to highway exit and soil quality are consistent with the results of the GATEs and SHAP values. The estimated coefficients for the parcel size and distance to the nearest highway exit are negative and statistically significant ($p < 0.01$), whereas soil quality shows no significant effect ($p = 0.79$).

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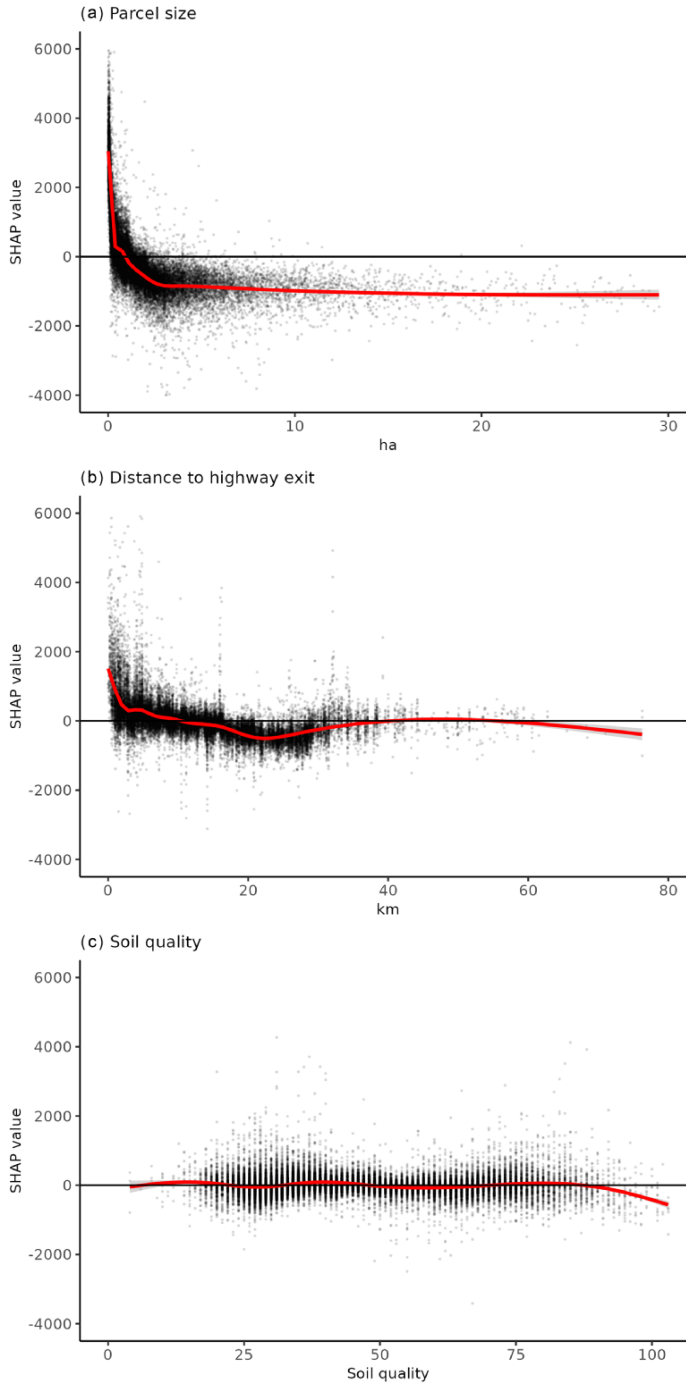

Fig. 5. SHAP values and Kernel smoother for (a) parcel size, (b) distance to highway exit and (c) soil quality.

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For the transaction-specific variables, year has a significant negative effect which means that the price markup of non-farmers erodes over time (–301 EUR/ha per year). This finding confirms previous results of converging prices by Curtiss *et al*. (2021) for the Czech farmland market. If land is sold by a non-farmer to a non-farmer, the price markup goes down by about 765 EUR/ha. Purchases for consolidation purposes also show a considerable impact on the CATE. A decline of the CATE can be expected if land is sold to a family member. Typically, prices are lower for these kinds of transactions, but the

Table 2. Results from the best linear projection

Variable	Estimate	Std. Error	t-value	Pr(>|t|)	
Plot-specific					
Parcel size (ha)	–124.15	43.60	–2.85	0.00	***
Soil quality (index)	2.41	9.07	0.27	0.79	
Grassland (D)	–582.24	293.26	–1.99	0.05	**
Distance to highway exit (km)	–30.26	13.11	–2.31	0.02	**
Transaction-specific					
Year (1 = 2005)	–301.49	84.18	–3.58	0.00	***
Seller type					
Non-farmer (D)	–765.75	351.52	–2.18	0.03	**
Public entity (D)	1,255.88	766.14	1.64	0.10	*
Private entity (D)	1,262.77	1,246.46	1.01	0.31	
Special circumstances					
Consolidation purchase (D)	4,087.54	606.30	6.74	0.00	***
Family relation (D)	–3,650.99	757.47	–4.82	0.00	***
Tenant buyer (D)	776.01	788.42	0.98	0.32	
Forced sale (D)	–2,388.79	1,922.23	–1.24	0.21	
Collector's price (D)	21,129.59	17,631.40	1.20	0.23	
Employment relation (D)	2,008.66	5,973.46	0.34	0.74	
Alternative land (D)	4,750.83	3,236.89	1.47	0.14	
County-specific					
Hog density (per 100 ha AA)	4.08	1.23	3.33	0.00	***
Cattle density (per 100 ha AA)	4.61	4.05	1.14	0.26	
Population density (1/sqkm)	4.57	6.21	0.74	0.46	
Population's average age	1,003.73	396.91	2.53	0.01	**
Net reproduction rate	383,242.44	250,882.21	1.53	0.13	
Migration rate	–82,516.47	41,724.29	–1.98	0.05	**
Agricultural area (share)	–1,536.83	1,884.12	–0.82	0.41	
Leisure area (share)	14,016.51	103,705.33	0.14	0.89	
Industrial area (share)	–53,626.87	32,405.27	–1.65	0.10	*
Street area (share)	–66,979.64	23,477.83	–2.85	0.00	***
Residential area (share)	62,235.06	28,503.65	2.18	0.03	**
Planned biomass (kW/ha)	–1,811.24	4,867.09	–0.37	0.71	
Planned wind energy (kW/ha)	–253.06	795.90	–0.32	0.75	
Constant	–38,567.29	16,667.15	–2.31	0.02	**

Note: D denotes a dummy variable; ***, ** and * denote statistical significance level at the 1 per cent, 5 per cent, 10 per cent significance level, respectively.

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discount appears to be more pronounced if it is sold to a non-farmer family member.

Among the county-specific covariates, the shares of residential area and street area, population's average age and hog density have a significant influence on the linear projection of the CATE. It is surprising to note that hog density has a positive impact, despite the fierce competition for land among farmers in Lower Saxony with intensive livestock production (Yang, Odening and Ritter, 2019). However, at the same time these regions are highly industrialised which may stimulate speculative demand among non-farmers. Although the positive coefficient of residential area indicates the effect of urbanisation on price mark ups, the negative impact by street area and the positive effect of the population's average age are conflicting.

7. Concluding remarks

In this paper, we scrutinise whether non-farmers pay more for farmland than farmers. This claim plays a pivotal role in the discussion about 'land grabbing' and 'fair' access to farmland. Non-farmers are often blamed for crowding out farmers from land markets due to their financial power. In view of this perceived threat, regulations have been suggested that grant farmers privileged access to land markets.

A reliable and empirically sound answer to this question is challenging because farmland prices are determined by a multitude of attributes. Thus, a straightforward comparison of average land prices paid by farmers and non-farmers may be misleading because different buyer types may self-select themselves into land transactions with different attributes. To address this problem and to determine price differentials that can be understood as 'treatment' effects of buyer types, we employ a causal forest approach. We apply this machine learning technique to a large data set of 35,016 individual land transactions in Lower Saxony, covering the period from 2005 until 2019. The first important finding is that a selection bias is present when simply comparing average price differences of farmers and non-farmers. While farmers pay, on average, 927 EUR/ha more for farmland, the ATE of non-farmers amounts to 1,998 EUR/ha (about 11 per cent). This implies that farmers prefer to buy land with attributes they value. Another important finding of our analysis is that treatment effects are heterogeneous rather than constant, i.e. they are moderated by certain land attributes. Specifically, our results indicate a decreasing trend in non-agricultural price premiums with increasing parcel size, leading to a negative impact of larger parcel sizes on these premiums. This is important because larger plots are more economically relevant for agricultural production than numerous, small land plots. At this point, one should recall that the decline of the CATE with plot size may, to some extent, reflect the fact that larger plots are subject to land market regulations that may prevent non-agricultural investors from either making bids according to their true WTP or from participating in the land market at all. However, given the rather low number of land market interventions through local authorities, we expect this potential source

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of bias to be small. In addition, our research shows that soil quality does not have a moderating effect on the price markup by non-farmers, suggesting that both farmers and non-farmers place a similar value on soil quality. This observation leads to the conclusion that regardless of the type of buyer, soil quality maintains a consistent valuation across transactions.

Generally, our results feed into the current discussion on tightening land market regulations. Our analysis confirms that non-farmers constitute an important group of land market participants. In our study region, over 40 per cent of the parcels have been bought by this group. Moreover, we provide evidence that non-farmers actually pay more, on average, for farmland with the same characteristics compared with farmers. Stakeholders or politicians following the dogma that farmland should be owned by farmers only (Bundesministerium für Ernährung und Landwirtschaft, 2021) may use these facts to call for more effective protection of farmers against financial investors. However, while we observe this substantial markup for the vast majority of transactions, these transactions only partially represent the land market in terms of the share of area sold. For larger parcels of land, price markups decrease significantly. Moreover, for these parcels farmers already have a right of first refusal under current land market regulation in Germany. For the largest parcels, 72 per cent of the transactions and 71 per cent of the transacted area are bought by farmers, which provides no evidence that farmers are driven out of farmland markets. Furthermore, the inflow of capital into the agricultural sector can be seen as an indicator of functioning capital markets and efficient factor allocation.

Knowledge about the heterogeneity of treatment effects is also potentially useful for the design of land market regulations that grant preferential market access to farmers. For example, attributes that moderate the price gap between farmers and non-farmers might be used to select land plots that are sold via restricted auctions in the context of land privatisation. While this type of land transaction constitutes a niche market in Germany (Hobe and Musshoff, 2021), it is of great importance for the privatisation process of formerly stated owned land in Eastern European counties (Kvartiuk, Herzfeld and Bukin, 2022).

From a methodological perspective, this paper showcases advanced causal machine learning approaches for the detection of heterogeneous treatment effects. It highlights the capability of causal forests in dealing with complex interactions and non-linear relationships in conjunction with large sets of observational data. Regarding the interpretation of the outcomes, the use of SHAP values reveals considerable individual heterogeneity of treatment effects, which is not visible from ATE and GATE. This heterogeneity, however, challenges the generalisability of covariates' impact on treatment effects and thus price markups of non-farmers.

The study can be extended in several ways. First, it would be interesting to replicate the results for other countries and time periods. In particular, it would be interesting to see if anything has changed with the new land market regulation in the study region. Second, other covariates should be investigated further, as suggested by the best linear projection. Third, it would be desirable

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to control for potential biases of land market policies on estimated price mark-ups. This could be achieved by conducting a closer examination of treatment effects at the cut-off point associated with plot size. This could be done through a regression discontinuity design, or an event study conducted before and after the implementation of new land market regulations that took effect in 2022 in Lower Saxony.

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Conflict of Interest

None declared.

Supplementary data

Supplementary data are available at *ERAE* online.

Appendix

A.1. Pseudo-treatments approach

To assess the unconfoundedness assumption, we apply a pseudo-treatment approach and test for non-zero causal effects of pseudo-treatments. This approach requires the availability of additional control information, specifically the existence of a two-component control group (Imbens and Rubin, 2015: 485–487). In our case, we have additional information if the buyer was a former tenant of the plot. Therefore, we divide the control group of our analysis, namely farmer buyers, into former tenants (pseudo-treatment group) and non-tenants (pseudo-control group).

Let G_i denote a generalised treatment group for unit i. This indicator variable takes, in our case, three values, $G_i \in \{c_{nt}, c_t, t\}$. Units with $G_i = c_t$ or c_{nt} (tenant or non-tenant) are subgroups of farmer buyers with $W_i = 0$, whereas units with $G_i = t$ are non-farmer buyers with $W_i = 1$:

$$W_i = \{ \begin{array}{c} 0 \text{if } G_i = c_{nt}, c_t \\ 1 \text{if } G_i = t \end{array} . \tag{A.1}$$

The crux of the unconfoundedness assumption is that the treatment assignment W_i is independent of potential outcomes $Y_i^{(0)}$ and $Y_i^{(1)}$ given covariates X_i, which is denoted as $\left(Y_i^{(0)}, Y_i^{(1)}\right) | X_i \perp W_i$. Since this is not directly testable with available data, we concentrate instead on assessing the implications of a stronger conditional independence relation, referred to as 'group unconfoundedness' (see Imbens and Rubin, 2015: 485–487):

$$\left(Y_i^{(0)}, Y_i^{(1)}\right) | X_i \perp G_i. \tag{A.2}$$

While this group unconfoundedness condition implies unconfoundedness, unlike the latter, it offers testable implications. We examine the implication that G_i is independent of

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$Y_i^{(0)}$ given X_i, when $G_i \in \{c_{nt}, c_t, t\}$. Since $G_i \in \{c_{nt}, c_t, t\}$ necessitates $W_i = 0$, the observed outcome Y_i^{obs} equals $Y_i^{(0)}$, leading to the conditional independence condition $G_i \perp Y_i^{\text{obs}} \mid X_i$ for $G_i \in \{c_{nt}, c_t, t\}$. Since group unconfoundedness is a stronger condition than unconfoundedness, it is crucial to understand if there are settings in which the weaker (and untestable) condition of unconfoundedness holds, but in which the stronger one (group unconfoundedness) does not. To scrutinise this possibility, we consider two alternative unconfoundedness-like conditional independence conditions, which are both entailed by the unconfoundedness assumption. If either of these conditions hold, then causal effects can be estimated by invoking the unconfoundedness assumption using only one of the control groups. It is, however, challenging to conceptualise a situation in which unconfoundedness, based on both control groups, holds true, but fails when considering just one control group. In practical terms, if unconfoundedness holds, the stronger condition is likely to hold as well, resulting in the testable implication concerning the conditional independence of G_i and Y_i^{obs} given X_i. In conclusion, this approach is based on the understanding that if stronger conditional independence assumptions are false, weaker conditional independence assumptions are likely also false. It is, however, important to note that while this methodology offers a plausible approach for assessing the unconfoundedness assumption, it is not infallible. The conclusions drawn from this approach depend on the plausibility of its underlying assumptions and the quality of the data.

Using additional information about the control group, we estimate the ATE between the two pseudo-treatment groups of tenants and non-tenants using AIPW with causal forests. The null hypothesis being tested is the following:

$$E\left[E\left[Y_i^{\text{obs}} | G_i = c_{nt}, X_i\right] - E\left[Y_i^{\text{obs}} | G_i = c_t, X_i\right]\right] = 0. \tag{A.3}$$

The left-hand side of the above equation represents the difference in the expected outcome Y_i^{obs} (the sales price) between non-tenants (c_{nt}) and tenants (c_t), conditional on covariates X_i. To estimate the ATE, we again use the method of AIPW with causal forest to control for X_i and different assignment probabilities. If the group unconfoundedness holds, we should not find a difference in the purchase price between tenants and non-tenants.

The estimated relative treatment effect amounts to 2.1 per cent with a confidence interval ($\alpha = 0.05$) of $[-4.5; +0.3\%]$. Hence, it is not significantly different from 0 and the null hypothesis cannot be rejected, suggesting that there is no significant difference in the purchase prices between the tenant and non-tenant groups, conditional on covariates. This provides evidence supporting the group unconfoundedness assumption, thereby indirectly supporting the credibility of the unconfoundedness assumption in our data.

A.2. Causal forest specifications and model calibration

The causal forest model is trained using the *grf* (generalised random forests) package in R, with the tune.parameters option set to 'all'. This means that all tunable parameters are optimised through cross-validation. The tunable parameters include 'sample.fraction', 'mtry', 'min.node.size', 'honesty.fraction', 'honesty.prune.leaves', alpha and 'imbalance.penalty'. The model was trained with 2,000 trees (num.trees = 2,000). The tuned parameters are shown in Supplementary Table A1. The calibration results are depicted in Supplementary Table A2.

A.3. Robustness checks

Our first approach (A) involves the replacement of our outcome variable Y with a placebo outcome, Y_{placebo}. Second, we substitute our treatment variable, W, with a random placebo

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variable, W_{placebo}(Approach (B)). In a correctly specified model, these placebo tests would yield minimal associations between the treatment and outcome variables. In a third step (C), we inject a random confounder (X_{random}) to our model to assess the stability of our baseline estimates, which should remain unaffected. This gauges the model's robustness against potential bias arising from unobserved confounding.

To evaluate the model's tolerance for unobserved heterogeneity bias, we exclude the most important covariate (Approach (D)). By leaving out the most important covariate and re-estimating the model, we assess the degree to which the correlation structure among the remaining observed confounders compensates for the absence of a key variable. This key variable is derived with the 'variable_importance' function from the *grf* package. The stability of the model under this condition suggests that it may be resistant to unobserved confounders that have not been included in the model, provided that these confounders are correlated with the observed set. Additionally, we do not only exclude one covariate but also exclude a group of covariates (Approach (E)) to simulate the situation in which a category of potential confounders is missing in the model. In our application, the group of county-specific variables is not considered. Again, in a robust model, we assume that due to the non-linear and interacting properties of the causal forest, the outcomes do not differ significantly from the original model.

Finally, we introduce a target common cause correlated with both treatment and outcome (Approach (F)), simulating various correlation structures between the target common cause, treatment and outcome. This robustness test is designed to indicate the model's sensitivity to potential hidden factors not captured in our latent covariate space, thereby giving us insights into its overall resilience and adaptability. The practical implementation of these robustness checks can be found in the code.

In the results of our robustness checks, our causal forest model exhibits impressive resilience (Figure A1). The placebo tests involving the replacement of the outcome variable Y with Y_{placebo} (A) and the substitution of the treatment variable W with W_{placebo} (B) indicate the model's robust specification. The associations between the placebo variables are minimal, conforming to our expectations in a well-specified model. Similarly, the introduction of a random confounder (X_{random}) into our model (C) does not considerably alter our baseline estimates, suggesting a high tolerance for potential bias from unobserved confounders. The results thus far affirm our model's capacity to maintain robustness in the face of potentially omitted variables. Our model further demonstrates substantial robustness when confronted with the exclusion of the most critical covariate, 'Parcel size' (D) and a set of covariates (E). The stable outcomes, even in the absence of these vital variables, testify to the causal forest's inherent strength in handling unobserved confounders that are correlated with the observed variables.

Finally, when we introduce a common cause (F) that is almost entirely dependent on outcome Y, it impacts the model's outcomes. However, the resulting manipulation in outcomes shows a relationship with the actual outcomes, thereby suggesting a controlled effect. Although this test stretches the model's resilience, it provides us valuable insights into its adaptability to handle potential hidden factors not accounted for in our covariate space. Overall, these findings endorse the model's robustness and strengthen our confidence in its validity and reliability.

A.4. Benchmark models

To evaluate the validity of the selected causal forest approach, we aim to modify not only the data set used in the study but also the method itself. To achieve this, we compare the results of the causal forest approach with three different estimation methods for assessing

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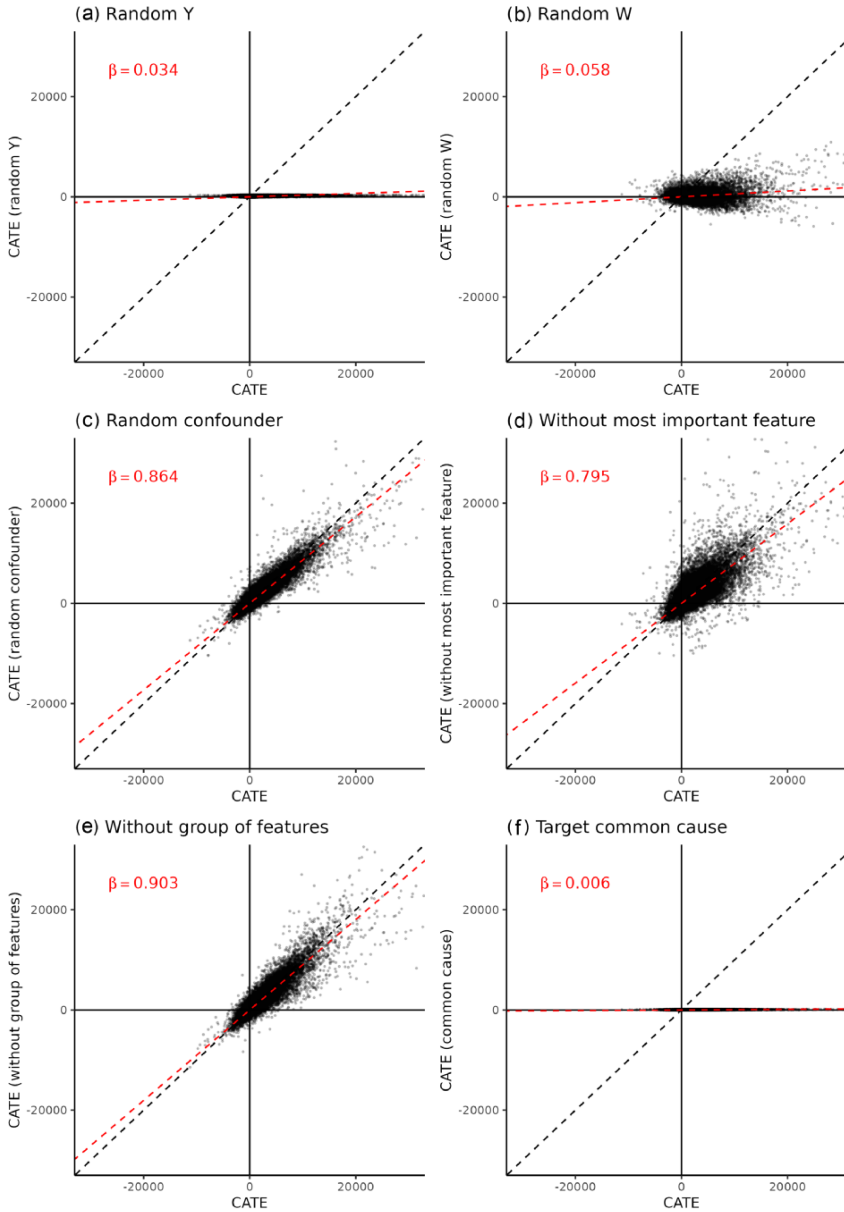

Fig. A1. Robustness checks outcomes for causal forest with all non-farmers. CATE is the conditional average treatment effect.

the impact of non-farmers on farmland sales prices. First, we estimate a hedonic regression model. The input variables X used are the same as those used for the causal forest model to ensure comparability. The treatment effect was modelled using a dummy variable W. W is set to 1 if the plot is purchased by a non-farmer and 0 if it is purchased by a farmer. Based on this approach, we estimate the following model where y is the price per ha using an OLS

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regression:

$$y = \beta X + \tau_{OLS} W + \varepsilon. \tag{A.4}$$

As a second benchmark model, we use the inverse propensity weighting approach (Austin and Stuart, 2015). The causal forest also uses propensity scores to estimate the ATEs and more specifically augmented inverse propensity weighting as explained in detail in the main part. In this method, the propensity score is estimated using logistic regression where the treatment variable, W, is regressed against the covariates X. The propensity score denoted as $e(X)$ represents the probability of a plot being purchased by a non-farmer given the covariates. In inverse propensity weighting, the propensity score is used to weigh the observations in estimating the ATE which can be written as displayed as:

$$\tau_{IPW} = \frac{1}{N} \sum_{N}^{i=1} \frac{Y_i . W_i}{\hat{e}^{-i}(X_i)} - \frac{Y_i . (1 - W_i)}{1 - \hat{e}^{-i}(X_i)} \tag{A.5}$$

Compared to the augmented inverse propensity weighting approach used in the causal forest, the inverse propensity weighting approach lacks doubly robust properties.

As a third benchmark model, we combine the first two benchmark models, namely inverse propensity weighting and outcome regression, using the same approach as in the causal forest, namely AIPW (Equation 10); however, the estimates were obtained by hedonic OLS regression and logistic regression.

Supplementary Table A3 shows that the treatment effects estimated with the benchmark models are slightly lower than those obtained with the causal forest method, but the difference is not substantial when looking at the overlapping confidence intervals. The confidence intervals of the benchmark models include the causal forest results. Furthermore, all methods result in a change in sign compared to the mean value comparison. The benchmark models result in lower precision than the causal forest in terms of confidence interval length. In total, the model achieved an R^2 of 0.43, which is lower than the R^2 for the causal forest (0.55). Using a logistic regression approach, a pseudo-R^2 of 0.14 can be achieved. This is once again a lower precision than the pseudo-R^2 of the causal forest (0.27).

Data availability

The transaction data used is not publicly available for confidentiality and privacy reasons, and we are not permitted to publish the original data set. If you wish to access the dataset for replication and verification purposes, we can provide access to the dataset in a secure environment. Due to the possibility of identifying individuals from the dataset, we are not allowed to share the data on external systems. Data sources are listed in the main text. To reproduce the results of the study, all calculations, tables and graphs can be reproduced using the code file 'code-non-farmers.txt' which contains the R-code used for this study. The code is also available in this online repository: https://github.com/lwschm/non-farmers. The code is structured in such a way that it can be executed and all calculations, tables and graphs used are exported as files. To use the code in different environments, make sure that the system is able to install the necessary R packages and adjust the paths used if necessary. If you would like to replicate our results from the paper and have access to the original data for replication purposes, we can provide a replication environment where it is possible to replicate the results with the original data sets and code. Please contact the corresponding author.

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